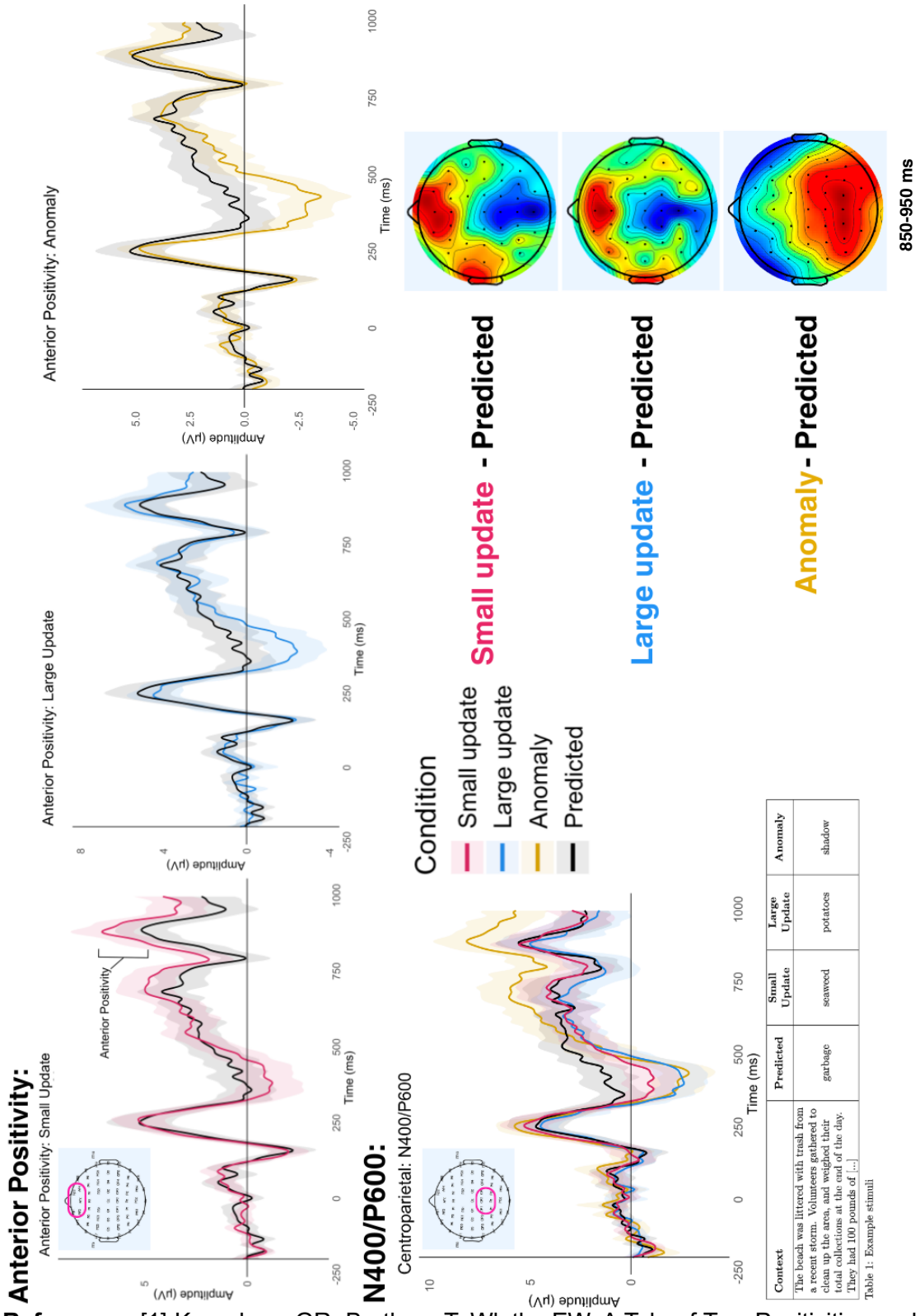
A Candidate ERP Index of Situation Model Updating During Language Comprehension

Background. Successful language comprehension involves building a situation model: a mental representation of the entities, events, and spatiotemporal structure of the meaning of discourse, based both on the incremental analysis of the bottom-up input, and on generalized event knowledge. When new input is encountered, the comprehender must update their situation model to accommodate the new information through a process that adjusts the model to integrate each new word's meaning. Despite the theoretical importance of this level of representation, the neural bases of situation model representations remain poorly understood. Although most theoretical conceptualizations treat the situation model as separable from the level of lexical-semantic processing, there is little neurophysiological evidence to support such a separation. Recent work has identified a late positive deflection of the ERP over anterior electrodes as a candidate index of updating the situation model[1] (anterior positivity; henceforth APos). If the APos reflects operations at the representational level of the situation model, as opposed to lower-level lexical-semantic access, then the amplitude of this ERP may be proportional to the *degree* of situation model updating required. We tested this hypothesis by comparing ERPs (N = 39) to discourse-embedded words that could be integrated into the situation model through either a small or a large update (see examples in **table** 1). Critically, both types of updates were *possible*, in the sense that they were both compatible with generalized event knowledge. We contrasted these two conditions with semantic anomalies, which could not be accommodated through an update based on generalized event knowledge. **Design.** Stimuli consisted of 120 three-sentence stories in English. In each story, the final sentence contained a target word which was either : 1) highly congruent with preceding context (**predicted**); 2) unexpected given the context, but plausible and relatively central to the event concept (**small update**); 3) unexpected given the linguistic context, but plausible and relatively *less* central to the event concept than the small update continuations (**large update**); 4) semantically anomalous (semantic **anomaly**). The two plausible continuation types were not significantly different in their GPT2-derived surprisal given the preceding context, nor in the cosine similarity between their Word2Vec embeddings and the predicted word. **Procedure.** The first two sentences of each story were presented at once and read in a self-paced manner, terminated by a button press. The final sentence of the story was presented in central RSVP (ISI = 550ms). After each story, participants indicated whether the story made sense by button press response. Participants were told in advance that some stories may be unusual, but should still be counted as making sense if they could come to an understanding of what the story meant, to encourage "doesn't make sense" responses only to the anomaly condition. Each target word was followed by 3-6 words of concluding context. **Analyses and Results.** We quantified three ERP components for analysis: **APos**: mean voltage 600-1000 ms post-word onset at anterior electrodes (Afz, AF3/4, FP1/2); **N400**: mean voltage 300-500 ms at centroparietal electrodes (Cz, Cpz, C1/2, CP1/2); **P600**: mean voltage 600-1000 ms at the centroparietal group. The **APos** was affected by continuation type (4-level ANOVA, $F(3,38) = 3.25$, $p < .05$), but only small update words elicited a larger **APos** than predicted in pairwise tests ($t(38) = 2.80$, $p < .01$). The **APos** to small-update words is consistent with the hypothesis that it is modulated by situation model updating. However, we did not confirm the prediction that **APos** would increase with the degree of situation model updating. All non-predicted words elicited larger **N400s** than predicted words (main effect: ($F(3, 38) = 10.87$, $p < .0001$); small update: $t(38) = 2.78$, $p < .01$; large update: $t(38) = 5.39$, $p < .0001$; anomaly: $t(38) = 4.06$, $p < .001$). The anomaly condition ERP was qualitatively different from the two update conditions in that it elicited the largest N400 and was the only condition to elicit a **P600** effect, relative to predicted ($t(38) = 4.18$, $p < .001$). This distinction in the ERPs indicates that the three unexpected word conditions should not be construed as lying along a continuum (small update < large update < anomaly) in terms of semantic unexpectedness.

Our failure to observe the predicted small < large update effect on the APos could indicate that APos is not related to situation model updating; however, we will conduct a follow-up study that refines the stimulus materials to achieve a better test of our original hypothesis.



Context	Predicted	Small Update	Large Update	Anomaly
The beach was littered with trash from a recent storm. Volunteers gathered to clean up the area, and weighed their total collections at the end of the day. They had 100 pounds of [...]	garbage	seaweed	potatoes	shadow

Table 1: Example stimuli

References [1] Kuperberg GR, Brothers T, Wlotko EW. A Tale of Two Positivities and the N400: Distinct Neural Signatures Are Evoked by Confirmed and Violated Predictions at Different Levels of Representation. J Cogn Neurosci. 2020 Jan;*32*(1):12-35. doi: 10.1162/jocn_a_01465. Epub 2019 Sep 3. PMID: 31479347; PMCID: PMC7299186.